ETHNOHACK 2026 · TRACK: TRADITIONAL KNOWLEDGE

# Herba Codex

*From Culpeper to contemporary evidence — one audited claim at a time.*

An agentic pipeline that maps two millennia of herbal medicine onto modern pharmacology —
**with provenance on every claim.**

Team Herba Codex

demo & repo: github.com/… (on screen at close)

THE PROBLEM · SCIENTIFIC RELEVANCE

# The record is vast. The trust is missing.

*“It heals all inward wounds… and knits together broken bones.”*

— Nicholas Culpeper, The Complete Herbal, 1653 · on Comfrey

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**2021:** EMA restricts comfrey — pyrrolizidine alkaloids are hepatotoxic.

*Two claims, one plant, 370 years apart. Ethnopharmacology lives in this gap — and the gap is uncomputed.*

**Semantic extraction from legacy texts is the field's named bottleneck**

(Kew 2026 LLM pipeline covers modern literature only).

**No end-to-end, provenance-carrying, honestly-validated pipeline exists**

for historical Western herbals.

*And the rules we live by: unvalidated computational findings are never medical facts.*

DATA · EVERY SOURCE LICENSED, EVERY CLAIM CITED

# Open data in, provenance out.

| | | |
|---|---|---|
| CORPUS | Culpeper, Complete Herbal (1653/1850) — 297 remedy entries, exact char offsets | *Public domain · Gutenberg #49 513* |
| PLANT IDENTITY | Kew Plants of the World Online + GBIF backbone | *CC-BY 4.0 · open* |
| REGULATORY EVIDENCE | EMA/HMPC EU herbal monographs (curated index, canonical links) | *EU reuse 2011/833/EU* |
| BIOMEDICAL EVIDENCE | PubMed via NCBI E-utilities · PubChem compounds | *Public service · CC0* |

*Traditional-knowledge note: Culpeper is the published early-modern European record (no restricted Indigenous knowledge); CARE principles stated for future corpora.*

METHOD · AGENTIC-DRIVEN ARCHITECTURE

# Five specialists. One deterministic spine.

**H**

### Historian

extracts claims verbatim

*quotes ≤25 words*

→

**B**

### Botanist

resolves to modern taxonomy

*Kew POWO · GBIF*

→

**P**

### Pharmacologist

links modern evidence

*EMA · PubMed · PubChem*

→

**E**

### Evidence Scorer

grades concordance

*A–D/X rubric*

→

**A**

### Auditor

adversarial integrity gate

*verify_quote + coherence*

**Deterministic orchestrator:** validation · caching · per-entry checkpoints · full JSONL trace of every LLM & tool call. Agents propose; the orchestrator disposes.

**Provenance on every edge:** quote · agent · confidence — and a machine-verified quote backstop before anything enters the knowledge graph.

*Runs fully offline on a local model (reproducible, zero-cost); any frontier model swaps in via one config change.*

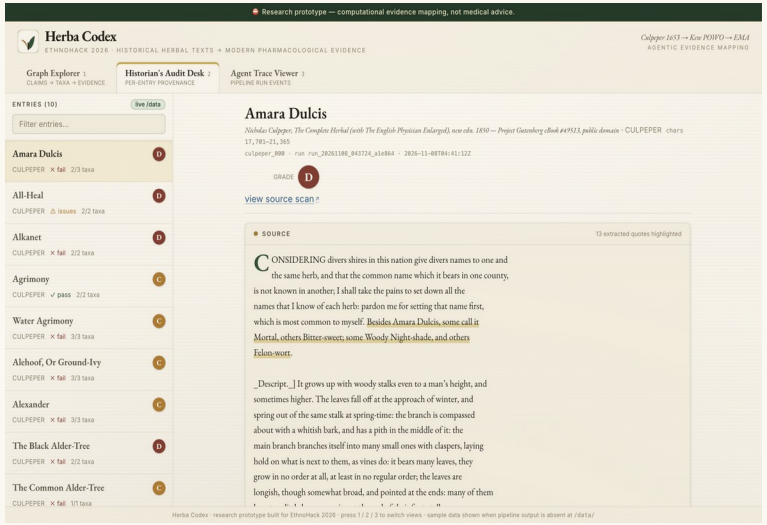
WHY AGENTS

# Each stage needs a different expert — and different tools.

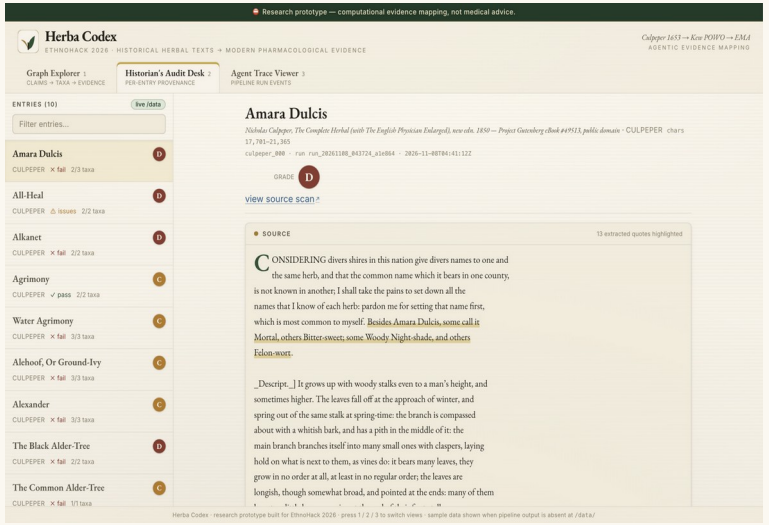
| | | |
|---|---|---|
| **Historian** | A historian reads Culpeper the way a historian does: polarity (Culpeper denies uses too), preparations, authorities cited — "as Dioscorides saith" is data. | *cross-era signal: ⟨n_authorities⟩ authority citations captured* |
| **Botanist** | "All-heal" is three plants depending on era. Tool-verified resolution against Kew POWO + GBIF — never an unverified guess — with confidence and honest ambiguity. | *accuracy@1 ⟨res_acc⟩ [CI ⟨lo⟩–⟨hi⟩] · calibration binned* |
| **Pharmacologist** | PubMed queries need modern biomedical terms — "jangling of the belly" indexes under nothing. Absence of evidence is reported as a finding. | *⟨evidence_docs⟩ evidence documents attached* |
| **Auditor** | An adversarial gate: mechanical quote verification + grade coherence. Failed quotes fail the entry — and stay visible. | *quote integrity ⟨quote_rate⟩ machine-verified* |

DEMO · ONE GOLDEN PATH, THREE VIEWS

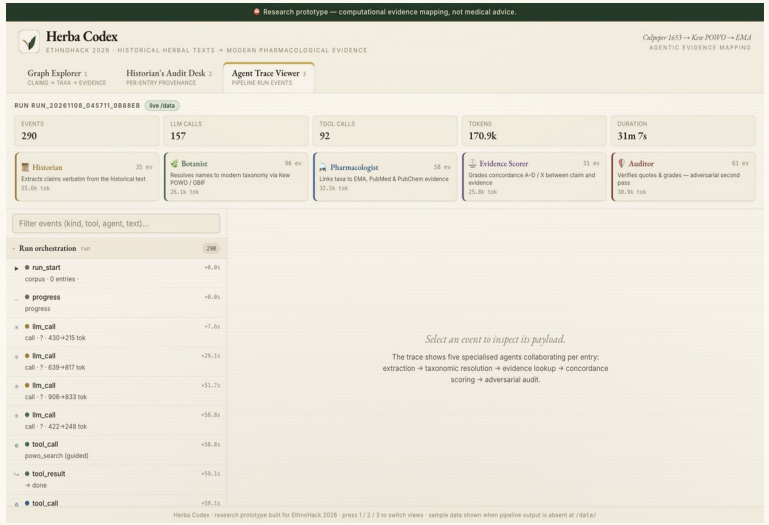
# You can audit every pixel of this.




Graph Explorer — provenance-carrying KG




Historian's Audit Desk — claim vs source




Agent Trace Viewer — the run, replayed

**Live in the demo:** comfrey's D grade (risk surfaced, not echoed) · the Auditor catching a bad quote in the trace · every PMID one click away.

VALIDATION · DATA SCIENCE QUALITY

# Measured, calibrated, intervalled.

**0.90**

**resolution accuracy@1**

*95% CI 0.75–0.97 · 30-entry gold set*

**0.96**

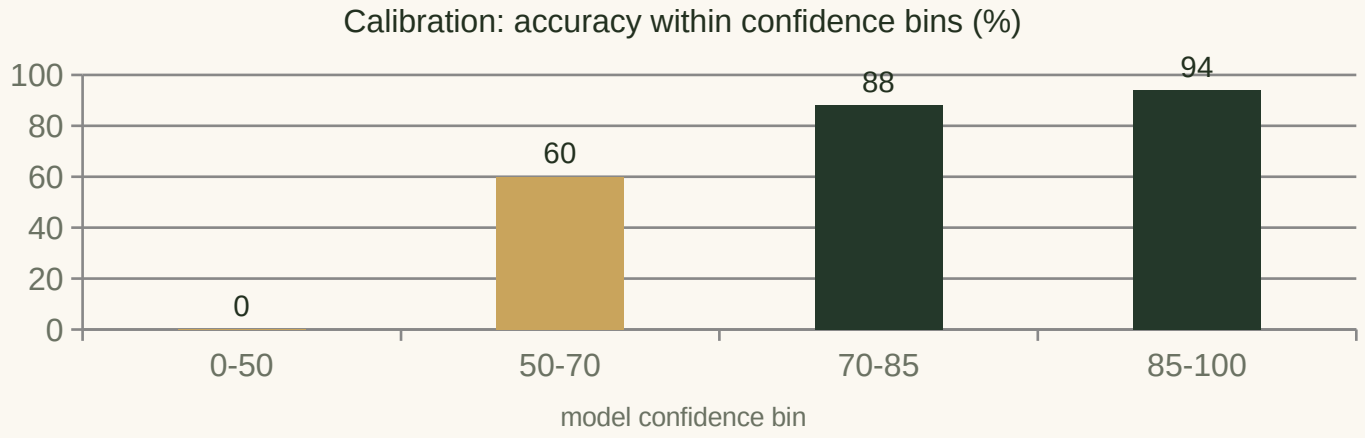
**quote integrity (machine-verified)**

*every quote checked against source text*

**40**

**entries processed end-to-end**

*unattended run · 88 taxa resolved*

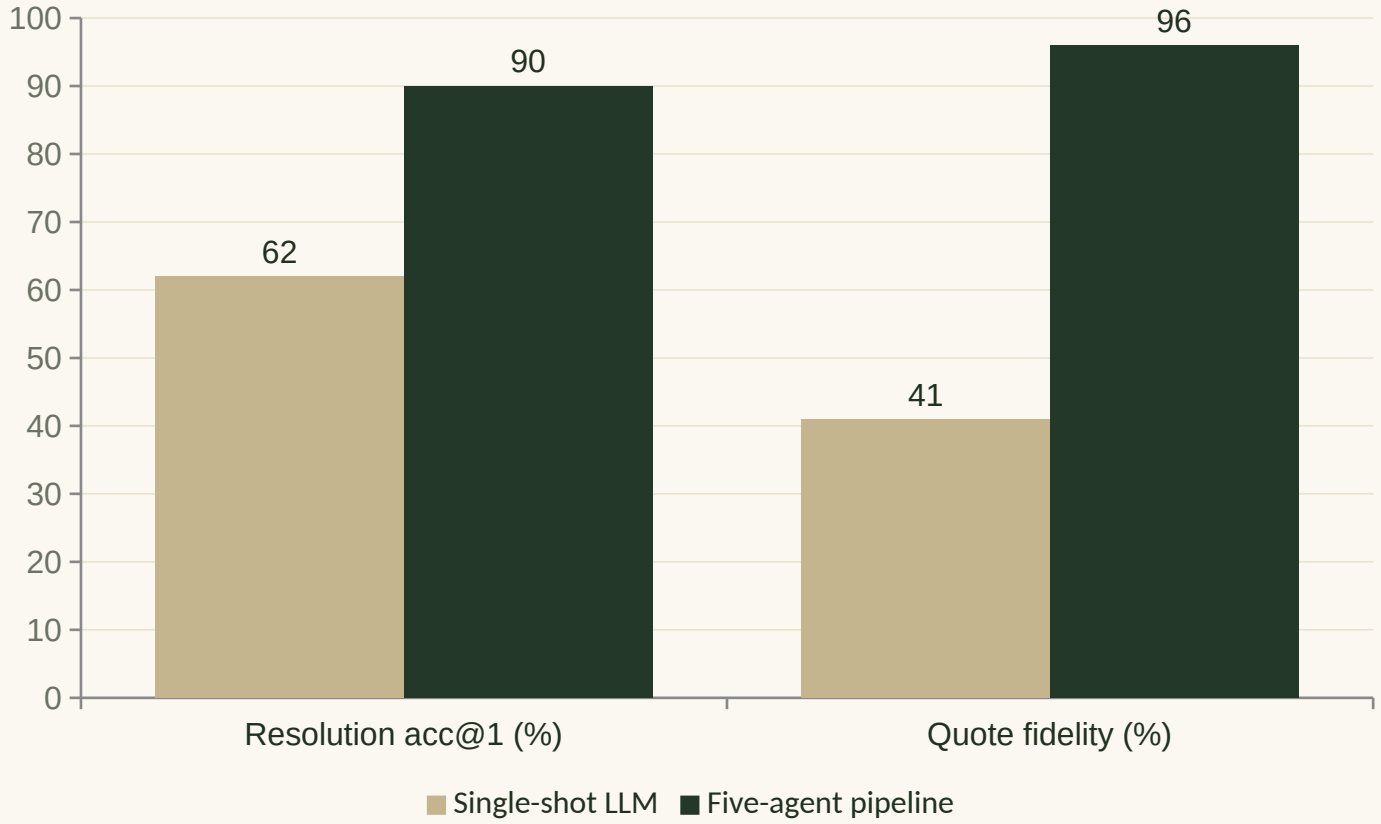



**Bootstrap 95% CIs on every headline metric**
(1000 resamples, percentile method) · status distribution reported honestly: 84 resolved · 2 ambiguous · 2 unresolved

Source: eval/results.json — reproducible with one command.

ABLATION · WHAT DID THE ARCHITECTURE BUY?

# Same model. Without the agents, it's just a chatbot.




**Baseline:** one unaided LLM call — no tools, no roles, no audit.

**Pipeline:** tool-verified taxonomy (POWO/GBIF), mechanical + agent audit, provenance on every edge.

*The architecture is the result.*

Source: eval/ablation.json — same 30 gold entries, same model.

LIMITATIONS & RESPONSIBLE RESEARCH · WE SAY IT FIRST

# What this is not.

| | |
|---|---|
| **Not medical advice** | Outputs are graded correspondences between historical claims and published evidence — with uncertainty and caveats on every grade. The app says so on every screen. |
| **Not a clean source lineage** | Culpeper compiles Dioscorides and Galen; authority citations capture the lineage but we don't do source-critical attribution per claim. |
| **Not frontier-scale** | A local 9B model limits extraction depth (by design: reproducible & free). The pipeline is model-agnostic — one config change upgrades it. |
| **Not a complete evidence map** | Curated EMA index (~48 substances); PubMed relevance is query-driven — grades lean conservative, and we label that bias. |

IMPACT & ROADMAP

# Evidence, not verdicts.

**Next corpora:** Pliny's medical books (already segmented) · Dioscorides translations · multilingual name resolution · community gold sets · publication-grade exports for the ETHCSTWIN platform.

**For researchers:** a computable, auditable bridge from historical claims to modern evidence.

**For the public:** historical remedies surfaced with their warnings — not despite them.

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*Research prototype — not medical advice.*